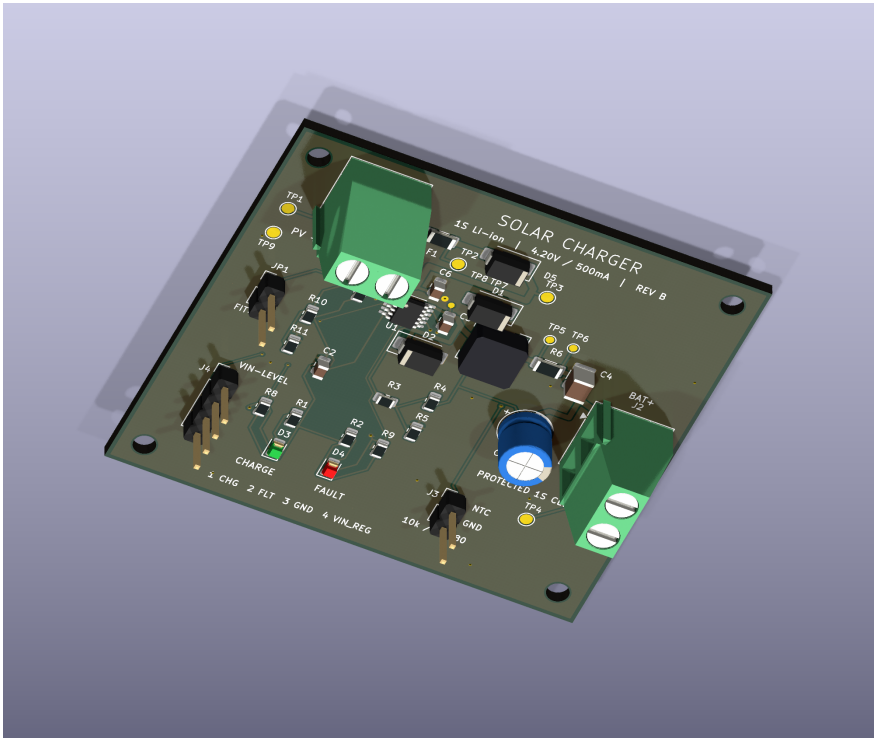
Solar Battery Charge Controller

Revision B — Final Design Report

Single-cell lithium-ion charging • LT3652 buck converter
4.2 V target • 500 mA nominal maximum • Two-layer PCB

Team member	Student ID	Primary contribution
Nishanth R	S20240020316	Schematic and circuit explanation
Siddarth Vijayan B	S20240020271	PCB planning and component placement
Arvind Kishor R	S20240020270	PCB routing, verification and fabrication preparation



Populated CAD assembly of Rev B. This is a render, not a photograph of assembled hardware.

Prepared for minor-project submission • 7 November 2026

Project status: CAD design and fabrication outputs prepared. Physical assembly, battery charging, LED operation, thermal performance and efficiency have not been measured. The latest native schematic requires KiCad 10 or newer.

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This report separates **design targets**, **recorded baseline checks**, **review of the submitted files**, and **proposed hardware measurements**. A calculated or expected value is not presented as a measured result.

Abbreviations: PCB — printed circuit board; CC/CV — constant current/constant voltage; ERC — electrical rules check; DRC — design rules check; NTC — negative-temperature-coefficient thermistor; BOM — bill of materials; SW — switching node.

1. Abstract, purpose and objectives

Revision B is a two-layer solar battery-charger PCB for one protected lithium-ion cell. An LT3652EMSE switching charger converts a higher solar-panel voltage into controlled CC/CV charging, with nominal settings of 4.2 V and 500 mA. The source is a nominal 12 V panel with maximum-power voltage near 18 V; its cold open-circuit voltage must not exceed 25 V.

The project develops an existing credited design into a documented PCB engineering prototype. It combines input protection, analog charging control, temperature sensing, status indication, accessible measurement points and fabrication outputs. Rev B adds a solar-input fuse, series reverse-polarity blocking diode, shutdown jumper and nine test points to the preceding revision.

Problem statement

A panel produces variable DC power and cannot directly enforce a lithium-ion cell’s charging limits. Direct connection can expose the cell to inappropriate current or voltage. A dedicated charger is needed to convert the input power and regulate the charge profile.

Objectives

• Implement the single-cell charging targets using fixed, documented component values.

• Develop readable schematic sections and a manufacturable two-layer PCB with project-local libraries.

• Add practical input-polarity blocking, shutdown access and labeled probe pads.

• Prepare a BOM, Gerber/drill outputs, 3D assembly, design-check evidence and staged bench-test plan.

Scope and intended applications

The board is intended as an educational solar-charging prototype and a possible charging subassembly for small battery-powered sensors, demonstration systems or low-power devices. A connected device may need additional output regulation, battery protection and load management. The board is not a 12 V battery charger, an inverter, a multi-cell charger or a complete solar power system.

Energy flow

The panel generates electricity; the battery stores energy. Weak sunlight reduces available charging power, so charging slows or stops. Battery discharge occurs when a load draws energy, with normal self-discharge and circuit leakage also possible. Low sunlight does not itself command battery discharge.

Development basis: The original solar-charger project, manufacturer documentation and KiCad libraries are credited in Section 17. CAD preparation and documentation used AI assistance; the team’s declared responsibilities are listed in Section 15.

2. Block diagram and design requirements

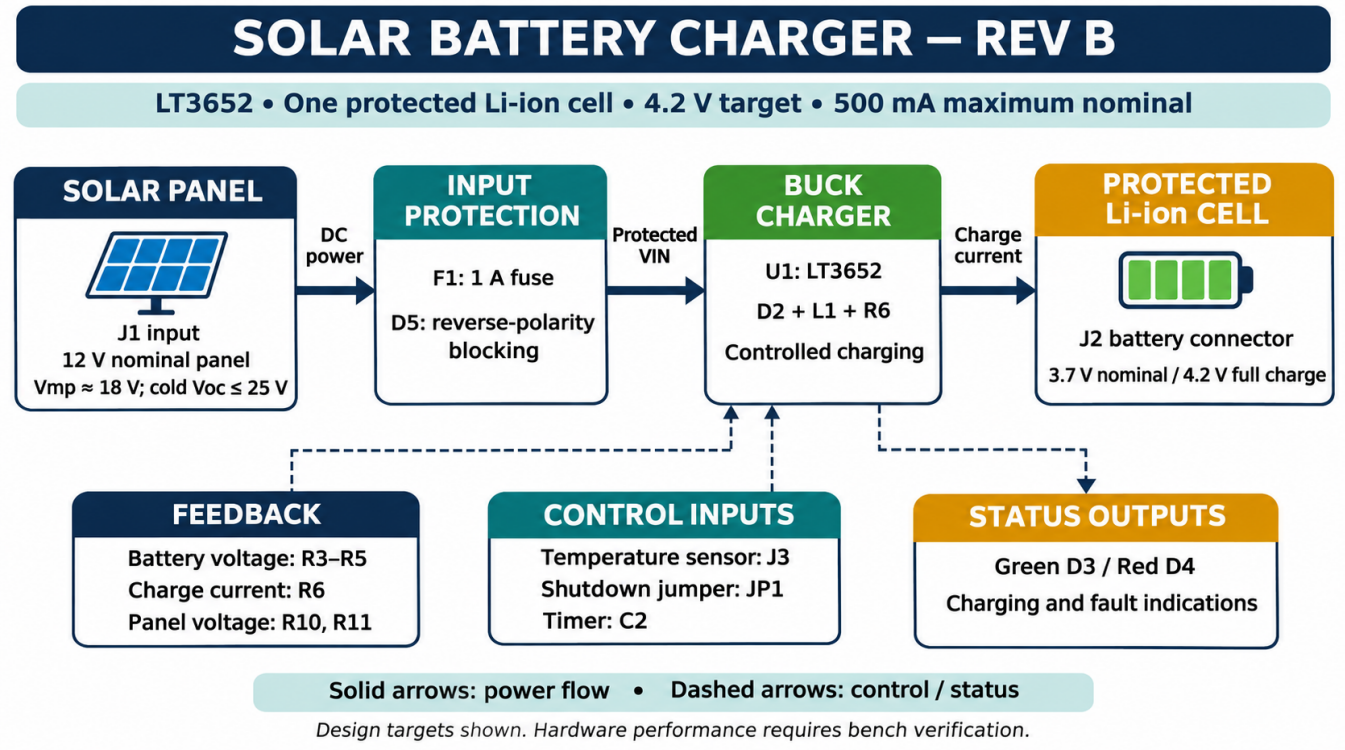


Figure 1. System block diagram. Solid arrows represent power transfer; dashed arrows represent feedback, control or status.

Parameter	Design requirement / target
Battery	One protected 3.7 V nominal lithium-ion cell; 4.2 V CC/CV chemistry; manufacturer permits ≥500 mA charging.
Charge voltage / current	4.200 V nominal / 500 mA nominal maximum; subject to component and IC tolerances.
Solar source	Nominal 12 V panel, Vmp approximately 18 V; at least 5 W recommended; cold Voc ≤25 V.
Input regulation	17.874 V nominal at protected VIN, after D5; panel-side voltage includes series drops.
Temperature sensing	External 10 kΩ at 25 °C NTC, B = 3380 K, attached to the cell; approximately 0–40 °C window.
PCB construction	60 × 55 mm; two copper layers; target 1.6 mm FR4 and 1 oz copper; four mounting holes.
Control and status	JP1 open permits charging; shunt disables. Green/red LEDs and a status/sense header.

The charger uses fixed input-voltage regulation to reduce loading when the panel voltage falls. It does not implement a continuously searching perturb-and-observe MPPT algorithm.

3. Schematic organization and power flow

The schematic contains eight functional sections. Matching net labels connect separately drawn points; they are not separate power sources. GND is the common return, not protective earth.

Section	Main parts	Purpose
1	J1, F1, D5, C1, C6, R7, JP1	Solar input, local bypassing, series protection and shutdown control.
2	U1, D1, D2, C3, L1, R6, C4, C5, J2	Buck power conversion, current sensing and battery connection.
3	R10, R11	Input-voltage divider connected to VIN_REG.
4	C2	Charging timer and associated fault supervision.
5	R3, R4, R5	Precision battery-voltage feedback.
6	J3 and external NTC	Battery-temperature monitoring.
7	D3, D4, R1, R2, R8, R9, J4	Charging/fault indication and status/sense access.
8	TP1–TP9	Input, output, current-sense, switching and ground probe access.

Main positive power path

J1 → F1 → D5 → U1 internal power switch → SW → L1 → R6 → BAT → J2. Battery negative connects to the common ground return. Feedback and timing branches measure or control operation; they do not carry all of the charging power.

Net	Meaning
VIN_RAW	Panel positive before the input fuse.
VIN_FUSED	Fuse output before D5.
VIN	Protected input after D5; supplies the charger and status circuitry.
SW	Rapidly switching connection between U1, L1, D2 and C3.
SENSE / BAT	Inductor and battery sides of R6; distinct nets joined through the sense resistor.
BOOST / VFB	Switch-driver bootstrap supply / battery-voltage feedback input.

PWR_FLAG symbols inform KiCad’s electrical checker that a net receives power; they are not physical components. Appendix A contains the full schematic reference export.

4. Input protection and charging power stage

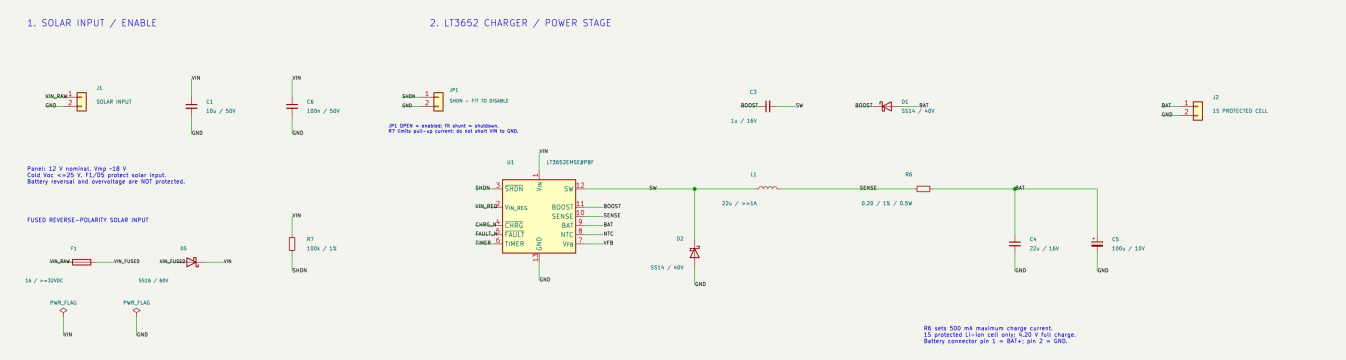


Figure 2. Schematic Sections 1 and 2. Use Appendix A for a larger, vector-quality view.

Solar input, bypassing and shutdown

J1 pin 1 supplies VIN_RAW. F1 is a 1 A soldered replaceable fuse rated at least 32 V DC; its interrupt rating and clearing curve must suit the actual source. D5 is an SS16-class, ≥60 V / 1 A SMA Schottky: anode towards VIN_FUSED, cathode towards VIN. Its series placement blocks reversed panel input, with a forward-drop power penalty. It does not protect against reversed battery wiring or input surges.

C1 (10 µF / 50 V) and C6 (100 nF / 50 V) bypass VIN locally. R7 (100 kΩ) pulls SHDN towards VIN. JP1 is open in normal operation; a shunt grounds SHDN and disables charging. Shutdown is not physical battery isolation.

Buck conversion and output

U1 switches at approximately 1 MHz. During the switch-on interval, input energy passes through the internal switch and L1 towards the battery. During the appropriate switch-off interval, D2 conducts from GND to SW so inductor current can continue. L1 is 22 µH, with a target saturation rating ≥1 A and RMS rating ≥0.5 A, subject to the selected part’s temperature limits.

R6 (0.20 Ω, 1%, 0.5 W) connects SENSE to BAT. U1 measures its differential voltage to regulate current. J2 pin 1 is BAT positive; pin 2 is GND. C4 (22 µF / 16 V ceramic) and C5 (100 µF / 10 V electrolytic) provide output capacitance. C5 polarity must be observed.

Bootstrap support

D1 is an SS14 diode with anode at BAT and cathode at BOOST. C3 (1 µF / 16 V) connects BOOST to SW. Together they support the internal high-side switch driver. The 16 V C3 rating applies across BOOST–SW, not across the full panel voltage.

> A protected cell is still required: the charging controller and the cell’s protection circuit perform different functions. Confirm actual part packages, effective ceramic capacitance and component ratings before purchasing or assembly.

5. Regulation, timer and temperature control

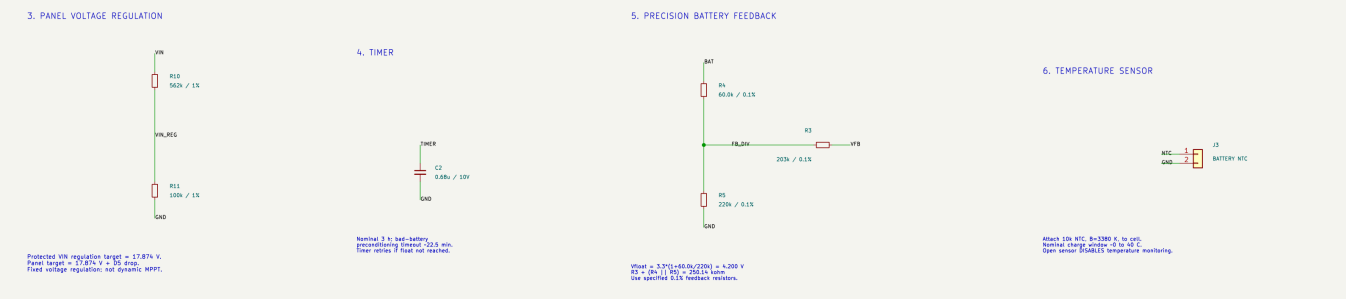


Figure 3. Schematic Sections 3–6: input regulation, timer, battery feedback and NTC connection.

Section 3 — input-voltage regulation

R10 (562 kΩ) and R11 (100 kΩ) divide protected VIN for VIN_REG. Near the nominal 17.874 V regulation threshold, U1 reduces charge demand to help prevent further input-voltage collapse. Because sensing is after D5, the corresponding panel voltage is higher by diode, fuse and wiring drops. The threshold does not guarantee operation exactly at the real panel’s maximum-power point.

Section 4 — timer supervision

C2 (0.68 µF / 10 V) sets a nominal 2.992 h timer interval, with associated preconditioning timeout near 22.4 minutes. A sufficiently low battery is first preconditioned at reduced current. Timer completion, restart and bad-battery response depend on the controller’s operating conditions; three hours is not a guaranteed charge duration.

Section 5 — battery feedback

R4 (60.0 kΩ) connects BAT to the divider junction; R5 (220 kΩ) connects the junction to GND. R3 (203 kΩ) connects the junction to VFB and provides bias compensation. All three are 0.1% parts. The nominal final-voltage target is 4.2 V.

Section 6 — temperature sensing

J3 connects an external 10 kΩ / B3380 NTC attached to the cell. Resistance decreases as temperature rises. The intended charging window is approximately 0–40 °C, subject to tolerances and hysteresis. An open sensor disables monitoring; it is not an open-sensor alarm.

Charging sequence

With suitable input, temperature and enable conditions, the charger preconditions a sufficiently low cell, then limits charge current up to the nominal 500 mA setting. Near 4.2 V it regulates voltage and current tapers. Timer and restart logic govern subsequent operation. Weak sunlight may limit current before the set maximum is reached.

6. Status indicators, connectors and test points

CHRG and FAULT are open-collector outputs. An asserted output sinks current through its LED branch from VIN. R1/R2 (10 kΩ) limit LED current; R8/R9 (100 kΩ) pull up the status nets.

Green D3	Red D4	Meaning with suitable protected input
On	Off	Charging at or above C/10, approximately 50 mA nominal.
Off	On	Bad-battery fault / preconditioning timeout.
On	On	Battery-temperature fault; charging paused.
Off	Off	May indicate standby, shutdown, insufficient input or current below C/10. It does not prove full charge.

External connectors

Connector	Pin connections and use
J1	1: panel positive / VIN_RAW; 2: GND.
J2	1: protected battery positive / BAT; 2: GND.
J3	1: external NTC; 2: GND. Attach the sensor to the battery.
J4	1: CHRG_N; 2: FAULT_N; 3: GND; 4: VIN_REG analog sensing node.

J4 limitation: Status pull-ups go to VIN, potentially near 25 V, and are not directly MCU-safe. Loading VIN_REG changes its divider ratio; use high-impedance measurement.

Probe locations

Point	Node / intended check
TP1 / TP2 / TP3	Raw panel input / fuse output / protected VIN after D5.
TP4	Battery-output voltage relative to GND.
TP5 / TP6	SENSE / BAT terminals of R6. $I \approx [V(TP5) - V(TP6)] / 0.20\ \Omega$.
TP7	SW switching activity; short probe connection required.
TP8 / TP9	Nearby scope ground / bench ground. Both are the same common GND net.

Measure across R6 in voltage mode; a current-meter input across the pads bypasses the resistor. Neither R6 terminal is GND. Switching measurements need suitable probes and verified instrument grounding.

7. Design calculations and component selection

The following values are nominal calculations from the selected component values and LT3652 relationships [3]. They are not measurements of assembled hardware.

Charge current: I = 0.100 / 0.20 = 0.500 A

R6 sets the current limit. IC sense-threshold tolerance and a 1% resistor give an approximate full-current expectation of 470–530 mA; insufficient input can reduce it further.

Sense-resistor dissipation: $P = I^2R = 0.5^2 \times 0.20 = 0.050$ W

The selected 0.5 W resistor provides nominal power margin; actual derating and thermal conditions must still be checked.

Final voltage: V = 3.3 × (1 + 60.0 / 220) = 4.200 V

R4/R5 set the ratio. R3 + (R4 ‖ R5) = 203 + 47.1429 = 250.1429 kΩ, close to the required 250 kΩ bias-compensation value.

Input regulation: V = 2.7 × (1 + 562 / 100) = 17.874 V

This is protected VIN after D5. Panel terminal voltage also includes the diode’s current/temperature-dependent drop and fuse/wiring drops.

Timer: $t = 0.68 \times 10^{-6} \times 4.4 \times 10^{6} = 2.992$ h

The associated preconditioning interval is approximately one eighth of this, about 22.4 minutes. Actual termination and restart follow the IC’s state conditions.

Inductor ripple estimate: ΔI ≈ 0.159 A peak-to-peak

Using Vout = 4.2 V, Vin = 25 V, f = 1 MHz and L = 22 µH: $\Delta I \approx V_{out}(1 - V_{out}/V_{in})/(fL)$. Peak current is about 0.580 A at 0.5 A output; the diode-drop-free estimate is approximate.

Output power at full-current target: P ≈ 4.2 × 0.5 = 2.10 W

Panel power must also cover losses and reduced illumination. A ≥5 W panel is recommended; no conversion-efficiency value has been measured.

Selection details

Choose C4 for at least 10 µF effective capacitance at 4.2 V after bias, temperature and tolerance effects. Verify C1’s effective input capacitance at the maximum operating voltage. D1/D2 are ≥1 A / 40 V Schottky-class parts; F1 and D5 require verification of the actual manufacturer ratings and footprints.

8. PCB component placement

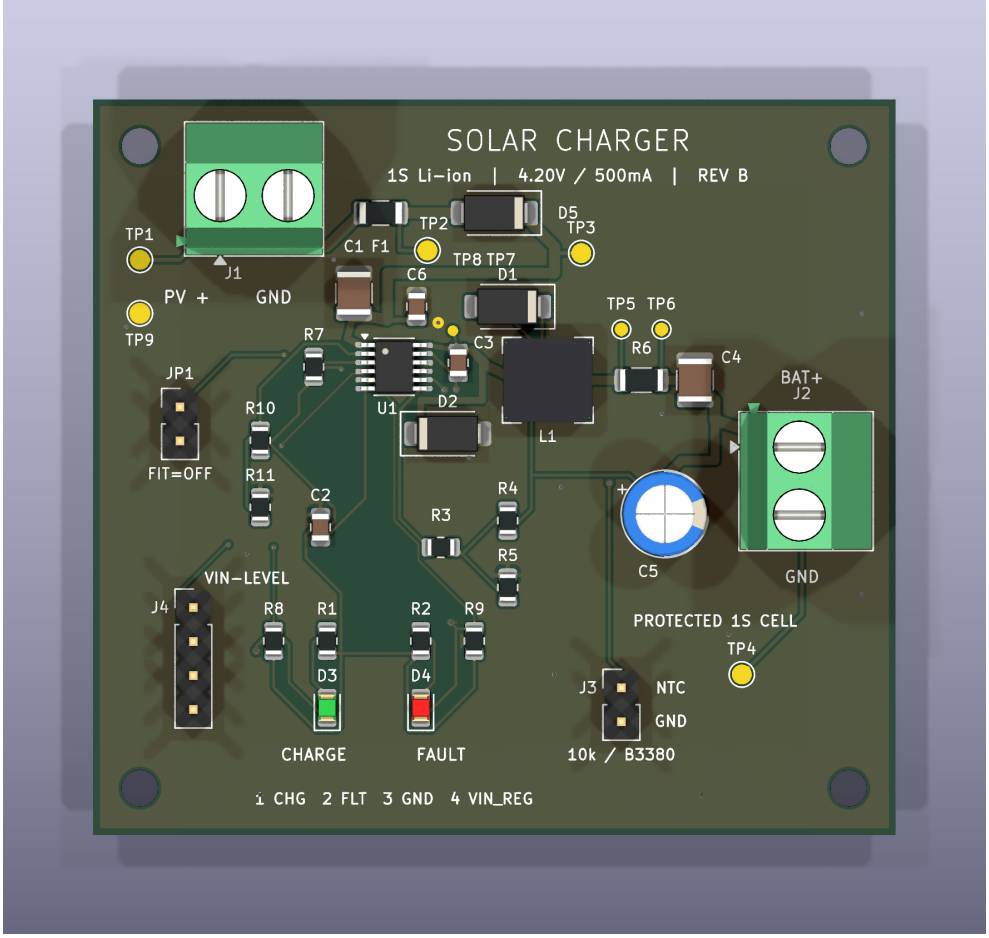


Figure 4. Populated top view. Positions are described in this orientation; the underside appears mirrored.

The layout uses a 60 × 55 mm two-layer board. J1 is near the upper-left edge, J2 is on the right, and the controller/power components occupy the central region. Status LEDs are near the lower edge; JP1 is accessible on the left and J3 near the lower right.

• **Input grouping:** F1 and D5 are placed near J1; local bypass capacitors C1/C6 serve the charger supply.

• **Power stage:** U1, D2, L1 and R6 are grouped to support short switching and charging connections. C4/C5 are near the output area.

• **Service access:** JP1, labeled probe pads and external connectors remain reachable. Four M2.5 clearance holes support mounting.

• **Assembly:** Most parts are SMD. Connectors and C5 are through-hole. U1 uses a 0.65 mm-pitch MSOP package with a grounded exposed pad; reflow and solder inspection are required.

Footprints must be checked against selected orderable parts, including pin numbering, polarity, body size, lead pitch and fuse interrupt/thermal ratings. Generic 3D fit is not a substitute for this check.

9. Copper routing, ground and thermal layout

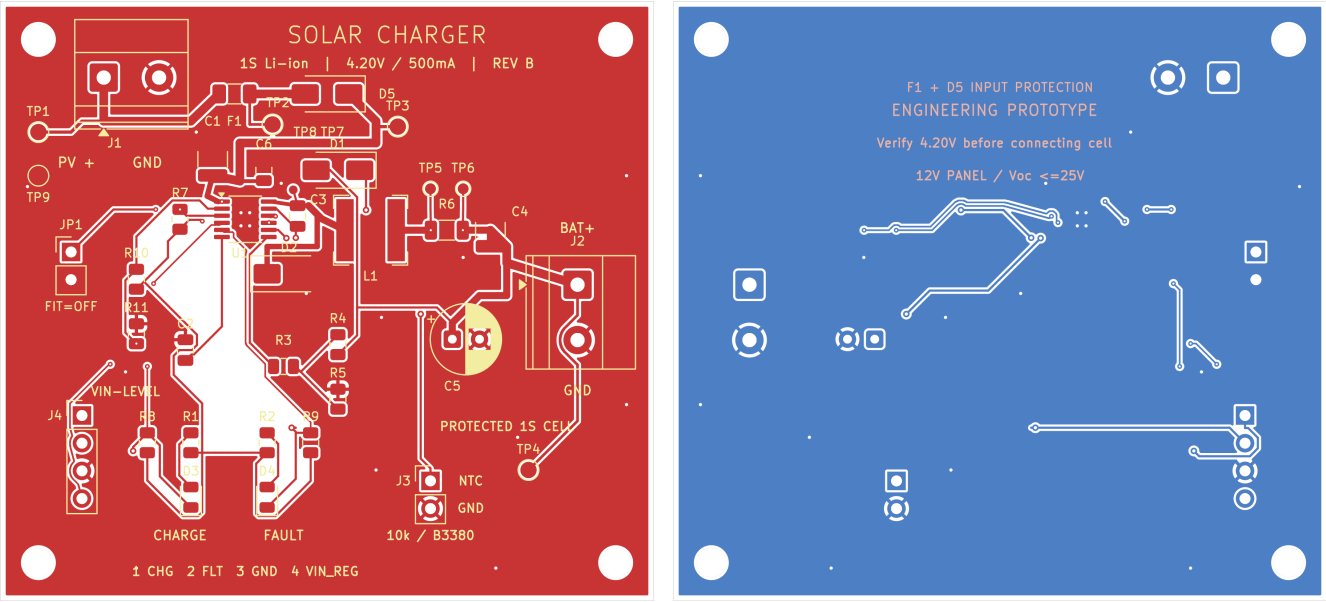


Figure 5. Front (left) and back (right) layer drawings. The native KiCad file remains the authoritative routing source.

Routing priorities

The input and output routes carry power, whereas feedback nets carry small sensing signals. Track widths and clearances must suit current, copper weight, voltage and permitted temperature rise. The design uses 0.15 mm minimum track/clearance geometry; smallest signal vias are 0.45 mm diameter with 0.20 mm drill, subject to fabricator acceptance.

The SW connection is a noisy switching node and should not be unnecessarily enlarged or routed near sensitive sensing. D2 and the input bypass return belong to compact switching loops. R6 sensing should sample its terminals without sharing avoidable high-current voltage drops. Feedback routes require controlled returns and separation from the switching node.

Ground and heat transfer

Ground copper and vias connect return paths on the two layers. U1’s exposed pad is connected to GND with four thermal vias to spread heat into the copper. This arrangement supports heat transfer but does not establish an acceptable temperature without measurement. Exposed-pad soldering and via solder-wicking need assembly-process review.

Probe layout

The SW probe uses a 0.8 mm pad and a short branch; the reviewed branch length is 1.504 mm. Nearby TP8 provides a short ground-return point. TP5/TP6 sit near R6 for differential voltage measurement. These pads are measurement stubs, not alternative high-current connection points.

10. Populated 3D assembly and portability

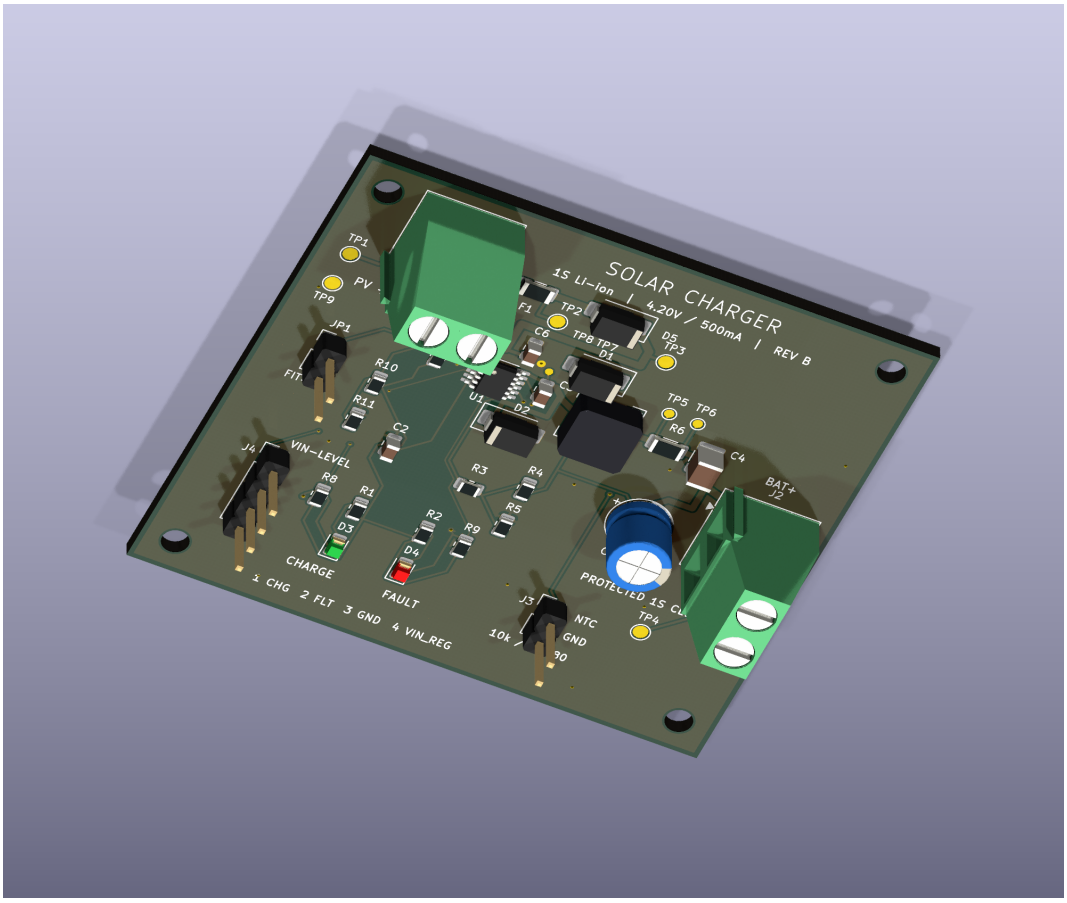


Figure 6. Angled CAD assembly. No panel, battery, thermistor, enclosure or live LED state is shown.

All 30 fitted components have local STEP model references. The four mounting holes and nine bare probe pads intentionally have no separate component bodies. JP1 is shown without a shunt, corresponding to the normal charging-enabled configuration.

Representation	Scope and limitation
Standard components	Generic KiCad package models, pinned source manifest; 16 bundled STEP files including color variants and unused-library support.
U1	Representative MSOP-12 model with 3 × 4.000 mm body versus nominal 3 × 4.039 mm. The 0.65 mm pitch is retained; not a vendor-qualified model.
L1	Original representative 6.6 × 6.2 × 2.8 mm envelope; not a verified vendor clearance model.
D3 / D4	Green/red body-color variants identify LED parts. They do not indicate emitted light or electrical operation.

Open the project in KiCad 10 or newer, launch PCB Editor and select View → 3D Viewer (Alt+3). Keep the 3dmodels folder beside the native files. The assembly STEP under Supporting documents/3D opens in FreeCAD. Mechanical tolerances and enclosure clearance must use actual part drawings.

11. Bill of materials

The table covers all 30 fitted board components. Probe pads and mounting holes are fabricated features, not purchased fitted parts. The separate BOM.csv includes detailed selection notes. No procurement cost is claimed.

Reference	Qty	Value / rating	Package	Function
U1	1	LT3652EMSE#PBF	MSOP-12 EP	Charging controller
J1, J2	2	2-pin, 5.08 mm	Terminal block	Panel / battery terminals
J3, JP1	2	2-pin, 2.54 mm	Pin header	NTC / shutdown access
J4	1	4-pin, 2.54 mm	Pin header	Status / VIN_REG access
C1	1	10 µF / 50 V	1210	Input bypass
C6	1	100 nF / 50 V	0805	High-frequency input bypass
C2	1	0.68 µF / 10 V	0805	Timer
C3	1	1 µF / 16 V	0805	Bootstrap, BOOST to SW
C4	1	22 µF / 16 V	1210	Output ceramic bypass
C5	1	100 µF / 10 V	Radial Ø6.3 mm	Output electrolytic; pitch 2.5 mm
D1, D2	2	SS14-class / 40 V	SMA	Bootstrap / freewheel
D5	1	SS16-class / ≥60 V	SMA	Solar-input polarity blocking
D3, D4	2	Green / red low-current LEDs	0805	Charging / fault indication
L1	1	22 µH; Isat ≥1 A	SRN6028	Power inductor; verify Irms/derating
R1, R2	2	10 kΩ / 1%	0805	LED current limiting
R3	1	203 kΩ / 0.1%	0805	Feedback bias compensation
R4	1	60.0 kΩ / 0.1%	0805	Feedback upper resistor
R5	1	220 kΩ / 0.1%	0805	Feedback lower resistor
R6	1	0.20 Ω / 1% / 0.5 W	1206	Charge-current sensing
R7, R8, R9	3	100 kΩ / 1%	0805	Shutdown / status pull-ups
R10	1	562 kΩ / 1%	0805	Input divider upper resistor
R11	1	100 kΩ / 1%	0805	Input divider lower resistor
F1	1	1 A / ≥32 V DC	1206 fuse	Fast acting; verify interrupt rating

External items: protected 18650 cell and suitable holder; nominal 12 V / ≥5 W panel; 10 kΩ / B3380 thermistor attached to the cell; suitable battery-lead fusing; rated wiring and a removable 2.54 mm shunt. Leave the shunt off JP1 for normal charging.

12. Verification evidence and current status

Recorded baseline results are from the prepared Rev B design on 6 November 2026 using KiCad 9.0.2. The submitted schematic was subsequently saved in KiCad 10. Baseline results must not be treated as a fresh full validation of that newer save.

Evidence	Result	Interpretation
Baseline ERC / DRC / parity	0 violations / 0 unconnected / 0 parity issues	Recorded in the original validation reports; earlier KiCad 9 revision.
Baseline topology / layout scripts	48 / 20 checks passed	Recorded baseline circuit-value and probe-layout checks.
Submitted PCB DRC, 7 Nov 2026	0 violations; 0 unconnected pads; 0 footprint errors	Fresh KiCad 9 PCB-only DRC. The uploaded PCB text matches the baseline apart from line endings.
Latest path review	30/30 component paths; both library paths resolve	Native PCB and libraries remain accessible in the reorganized package.
Latest independent layout review	19 checks passed; final strict severity check failed	Geometry/probe/exclusion checks passed. Four ERC rule categories are set to ignore in the KiCad 10 project.
Latest schematic ERC / parity	Not rerun in this environment	KiCad 9 cannot load the submitted KiCad 10 schematic. Use KiCad 10 for final ERC/parity.
Hardware / electrical simulation	Not performed / not included	No measured charging, LED, efficiency, temperature or simulation result is claimed.

ERC settings requiring review

The latest project lists **footprint_filter**, **four_way_junction**, **simulation_model_issue** and **single_global_label** as ignored ERC categories. These can be version/default dependent; the presence of an ignored category is not itself proof of a wiring fault. Review applicability, document the settings, and rerun ERC in KiCad 10. No per-item ERC or DRC exclusions were found in the reviewed project.

Final review action

Run ERC and schematic/PCB parity in KiCad 10, review all errors, warnings and ignored categories, and save dated reports. The supporting folders were reorganized; some original scripts and document links require path updates. The geometry-check script was evaluated against the actual project root during this report review.

13. Proposed hardware verification plan

The board is not yet assembled or physically tested. The sequence below is a planned verification process, not a record of completed measurements.

• **Before power:** verify part numbers, footprints, diode/LED/electrolytic polarity, exposed-pad soldering, fuse continuity and absence of unintended input/output shorts.

• **First startup without a cell:** use a current-limited bench source with JP1 disabling charging; check raw, fused and protected input voltages and unexpected current/heating.

• **Controlled charging test:** enable only after startup review; use a battery emulator capable of sinking current or a suitable electronic load in CV mode. A source-only supply is not a battery emulator.

• **Control/fault tests:** compare JP1 states, NTC normal/hot/cold conditions, CC/CV taper, input-voltage regulation, timer and restart behaviour. Record actual conditions and waveforms.

• **Real cell last:** use the specified protected cell, correct polarity, attached NTC, appropriate battery-side fusing and an attended test after the controlled tests pass.

Check	Expected design behaviour	Measured status
Protected VIN	TP3 = TP2 minus actual D5 drop	Not measured
Charging output	Approximately 4.20 V while regulating	Not measured
Charging current	Nominal maximum 500 mA with adequate input	Not measured
R6 differential voltage	Approximately 100 mV at 500 mA	Not measured
Shutdown	JP1 closed pulls SHDN low; charging disabled	Not measured
Temperature fault	Suitable hot/cold NTC conditions pause charge; both LEDs on	Not measured
Input regulation	Charge demand reduces near TP3 ≈17.874 V	Not measured
Taper / timer / temperatures	Within controller and actual component/cell requirements	Not measured

Use voltage mode across TP5/TP6; do not connect a current-meter input across a supply or battery. Do not attach an earth-referenced scope ground to either R6 terminal. Do not deliberately short the board to demonstrate the fuse. Full staged instructions are in TEST_PLAN.md.

14. Revision comparison, limitations and future work

Aspect	Rev A	Rev B
Charge targets	4.2 V / 500 mA nominal	Unchanged
Charging/fault LEDs	Present	Retained
Solar-input fuse	No onboard F1	Adds F1, 1 A / ≥32 V DC
Input reverse-polarity blocking	No added D5 input stage	Adds SS16-class series diode
Shutdown access	Pull-up enables controller	Adds JP1 shunt-to-disable header
Dedicated probe pads	No Rev B probe set	Adds TP1–TP9
Status-header labeling	Earlier SET label	Corrected VIN_REG label

Remaining limitations

Limitation	Practical implication
Open NTC disables monitoring	Sensor attachment and continuity need checks; add fail-safe sensing in a future revision.
No battery-reversal / surge protection	Observe battery polarity and ≤25 V cold panel Voc; additional protection requires design review.
Panel-limited fuse clearing	A low-current panel may not clear F1 during a short; assess actual source and fuse curves.
Fixed input threshold	The setting is not adaptive maximum-power tracking across all panel conditions.
No separate load management	External loads need review; their current can affect charging/termination behaviour.
VIN-level status / unmeasured performance	Use level conversion for logic interfaces. Efficiency, thermals, EMC and real charge operation remain unverified.

Next development steps

Complete final KiCad 10 checks, verify actual purchasable parts, build one board and collect staged measurements. Later work can evaluate battery-reversal/surge protection, open-sensor detection, logic-safe status outputs and a dedicated load path. Changes require recalculation and renewed verification.

15. Individual contributions

The following distribution records the team responsibilities supplied for this submission. It separates schematic work, component placement, and routing/verification. It does not assign fabricated hours or imply that hardware testing has been completed.

Nishanth R — S20240020316

Primary responsibility: schematic design, circuit study and explanation.

• Study the credited starting circuit and LT3652 functions; organize the schematic into its eight sections.

• Explain solar-input protection, buck operation, CC/CV charging, feedback, temperature sensing, timer and LED status.

• Review the 4.2 V and 500 mA setting calculations, connector functions and Rev A-to-Rev B changes.

Presentation evidence: block diagram, schematic walkthrough, component-function explanations and design calculations.

Siddarth Vijayan B — S20240020271

Primary responsibility: PCB planning, footprint review and component placement.

• Review board dimensions, component packages, pin numbering and polarity requirements.

• Explain input/output connector placement and the grouping of U1, bypass capacitors, diode, inductor and sense resistor.

• Review access to JP1, the NTC connector, LEDs, probe pads and four mounting holes; use the top/3D views to explain arrangement.

Presentation evidence: footprint and placement discussion, top-board walkthrough and assembly-access review.

Arvind Kishor R — S20240020270

Primary responsibility: PCB routing, design verification and fabrication preparation.

• Explain power and signal routes, current-sense/feedback considerations, ground copper, vias and exposed-pad thermal connections.

• Review recorded ERC/DRC evidence and the latest-file limitations; identify the final KiCad 10 review actions.

• Explain front/back layers, Gerber and drill outputs, BOM usage and the proposed hardware-test sequence.

Presentation evidence: layer/routing walkthrough, validation discussion, fabrication-output review and final 3D inspection.

Shared responsibility: understand the full design, preserve original-source and library credits, review the report and present each assigned section. CAD support, 3D integration and report preparation used AI assistance; team roles do not establish independent physical-test evidence.

16. Tools, deliverables and conclusion

Tool / resource	Use
KiCad	Schematic, PCB, library management, baseline ERC/DRC, Gerber/drill export and 3D viewing. Latest schematic needs version 10+.
FreeCAD / STEP	Standalone mechanical viewing; recorded model and assembly import checks.
KiCad libraries	Project-local symbols, footprints and generic component models, with notices retained.
Verification scripts	Supplementary topology, value, probe-layout and path checks; scripts may need updated paths after reorganization.
AI assistance	Support for CAD preparation, model integration, calculations, scripts and documentation; not a substitute for student review or physical testing.

Submission files

Item	Purpose
Three native KiCad files	Editable project, schematic and PCB.
Local libraries / tables / 3dmodels	Resolve component definitions, footprints and populated 3D bodies. Keep beside the native files.
BOM.csv and README.md	Parts/rating selection notes and project-opening instructions.
Supporting documents/manufacturing	Copper, mask, paste, silkscreen, outline Gerbers and drill output.
Supporting documents/previews and 3D	Schematic/assembly drawings, PNG/SVG board views and STEP assembly.
Validation / test plan / attribution / licenses	Recorded checks, planned measurements, original sources and library notices.

Conclusion

Rev B provides a documented PCB design for a buck-based solar charger targeting one protected lithium-ion cell at 4.2 V and up to 500 mA nominal. It extends Rev A with an input fuse, series polarity-blocking diode, shutdown jumper, labeled test points and clearer status/sense labeling.

The delivered CAD, model and fabrication files support design review and future assembly. The latest PCB passes the reviewed PCB-only DRC, while final KiCad 10 ERC/parity and rule-setting review remain to be completed. Reliable claims about charging, temperature, efficiency and protection require assembly and the planned bench tests.

17. References and attribution

1. Team project repository

https://github.com/nishanthr24-proj/PCB-desgning-

Editable Rev B project and supporting artifacts. Source materials reviewed for this report are the latest user-supplied ZIP and its recorded validation files.

2. Original solar charger

https://github.com/Hirusha785/solar-battery-charge-controller-pcb

Starting design supplied by the user; its README credits Fernando S.M.H.G. The supplied source archive identifier is 8adcedc8ef8a8e4cd02e30468040b93200b75584. Rev B is a derivative revision, not a claim of original invention of the LT3652 topology.

3. Analog Devices / Linear Technology, LT3652 datasheet

https://www.analog.com/media/en/technical-documentation/data-sheets/3652fe.pdf

Reference 3652fe: charging operation, pin functions, component selection, input-voltage regulation, timer, status decoding and layout. Verify the latest manufacturer revision and orderable package before assembly.

4. KiCad documentation

https://docs.kicad.org/

Schematic/PCB editing, libraries, electrical and PCB rule checks, fabrication outputs and 3D Viewer.

5. KiCad component-model library

https://gitlab.com/kicad/libraries/kicad-packages3D

Bundled generic STEP sources pinned to b8b3cfdfad88ba66f21002b3de51dc6f7d55ba5a. CC BY-SA 4.0 with the electronic-design exception; per-model sources are in 3dmodels/SOURCES.json.

6. SparkFun Sunny Buddy

https://github.com/sparkfun/SunnyBuddy

Consulted for LT3652 circuit lineage; its Eagle geometry, artwork and logos were not copied into this PCB.

7. FreeCAD documentation

https://www.freecad.org/

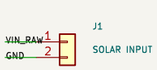
Mechanical viewing of the exported STEP assembly; no electrical simulation is implied.

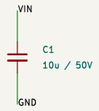
The user-supplied DC-DC-Buck-converter-main.zip inspired the populated front/back presentation. Its README credits Nandeesh K S; no reference-board geometry or artwork was copied. Existing library notices remain applicable. No new license is asserted for upstream work.

Appendix A — full schematic reference sheet

The following A3 landscape page reproduces the supplied vector schematic export for readable zoom/print review. It is the recorded Rev B reference drawing; the submitted native schematic was subsequently saved in KiCad 10. Use the native file for final ERC and schematic/PCB parity checks.

1. SOLAR INPUT / ENABLE

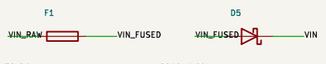


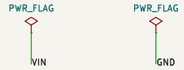




Panel: 12 V nominal, Vmp ~18 V
Cold Voc <=25 V. F1/D5 protect solar input.
Battery reversal and overvoltage are NOT protected.

FUSED REVERSE-POLARITY SOLAR INPUT







2. LT3652 CHARGER / POWER STAGE

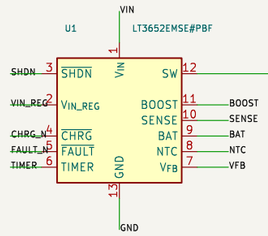


JP1 OPEN = enabled; fit shunt = shutdown.
R7 limits pull-up current; do not short VIN to GND.

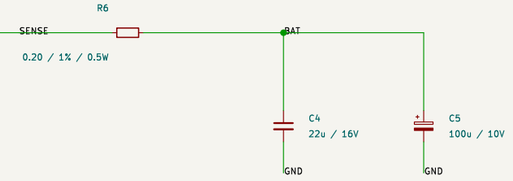












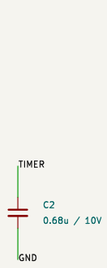
R6 sets 500 mA maximum charge current.
1S protected Li-ion cell only: 4.20 V full charge.
Battery connector pin 1 = BAT+; pin 2 = GND.

3. PANEL VOLTAGE REGULATION



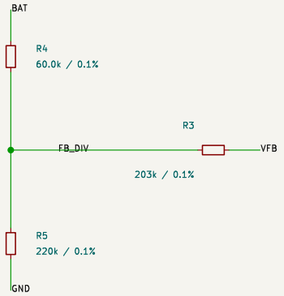
Protected VIN regulation target = 17.874 V.
Panel target = 17.874 V + D5 drop.
Fixed voltage regulation; not dynamic MPPT.

4. TIMER



Nominal 3 h; bad-battery preconditioning timeout ~22.5 min.
Timer retries if float not reached.

5. PRECISION BATTERY FEEDBACK



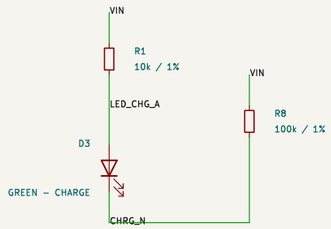
Vfloat = 3.3*(1+60.0k/220k) = 4.200 V
R3 + (R4 || R5) = 250.14 kohm
Use specified 0.1% feedback resistors.

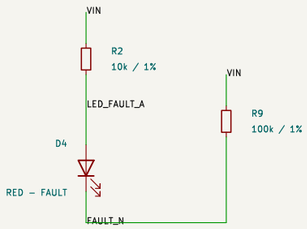
6. TEMPERATURE SENSOR



Attach 10k NTC, B=3380 K, to cell.
Nominal charge window ~0 to 40 C.
Open sensor DISABLES temperature monitoring.

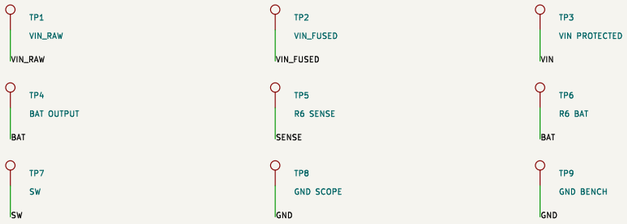
7. ADDED STATUS LEDs





Green only: charging >=C/10 (50 mA nominal).
Red only: bad-battery timeout. Both: NTC fault.
Both off: standby, no input, or low charging current.
LEDs are powered from solar VIN; off does not prove full charge.

8. REV B PROBE POINTS (see test plan)





VIN-LEVEL STATUS
NOT MCU SAFE
VIN_REG: analog sense only

ENGINEERING PROTOTYPE - ERC/DRC do not replace bench verification. No lithium cell until dummy-load checks pass.

Based on the uploaded Fernando S.M.H.G. project; corrected circuit and layout
1S Li-ion / 4.20 V / 500 mA - engineering prototype

Sheet: /
File: Solar_Charger_LED_RevB.kicad_sch

Title: Solar Charger Rev B - Fused Input, Shutdown and Test Points

Size: A3	Date: 2026-11-06	**Rev: B**
KiCad E.D.A. 9.0.2		Id: 1/1